

Name: _____

Class: _____

Mr. Rawson • GCSE Computer Science



The Byte Strip

Eight cards. No pens, no arithmetic, no subtraction.

Start here. Cut the eight place-value cards off the **card sheet**. Lay them on the mat below, **biggest on the left**, one card in each white cell.

The back of every card is blank, and that is the point. A card **face up** means that place value is switched **on** — a 1. A card turned **blank side up** means **off** — a 0. The grey number *above* each cell tells you which card it is, so you never lose your place.

One binary number. We are going to read it two ways.

00110100

Part A • Binary to denary

two steps

Step 1 Set the cards to match the number. **Face up** for a 1, **blank side up** for a 0. Work left to right.

Step 2 **Add up the place values that are face up.** Nothing else counts, and nothing is subtracted.

128	64	32	16	8	4	2	1

$32 + 16 + 4 = 52$
So **00110100** is **52** in denary.

Now you. Set each one on the mat and add up what is face up.

01101010 = _____

10000011 = _____

Part B • The same number, in hexadecimal

four steps — and the number has not changed

Name: _____

Step 1 Split the byte into two nibbles. A nibble is four bits. Slide the four right-hand cards away from the four left-hand cards.

0011 0100

Step 2 Work out the denary value of each nibble, using the place values for a four-bit number. Read each mat on its own — on its own mat a nibble reads 8 4 2 1, whichever half of the byte it came from.

8	4	2	1

$$2 + 1 = 3$$

the gap

8	4	2	1

$$4 = 4$$

Step 3 Change each denary value into its hex digit. Nought to nine are the same in both. Ten to fifteen become the letters A to F.

$$3 \rightarrow 3$$

$$4 \rightarrow 4$$

Step 4 Read the hex digits from left to right.

$$00110100_2 = 34_{16}$$

And it is still 52. 34 in hex *is* 52 in denary — check it: $3 \times 16 + 4 = 52$. One pattern of eight cards, written down two ways. **Hexadecimal is just where you put the gap.**

Your reference strip • build it, keep it

Use **one** nibble mat. Make every number from 0 to 15 with your four cards, and fill in the two blank rows. **Denary along the top, hex underneath** — one column is done for you in each strip.

Denary	0	1	2	3	4	5	6	7
Hex	0							
Binary	0000							

Denary	8	9	10	11	12	13	14	15
Hex			A					
Binary			1010					

Four cards run out at 15 — the whole reason hex needs letters.

Now turn over.

Part C • Hexadecimal back to denary

On page 2 you split 0011 0100 into two nibbles and read **3** and **4**. But the left nibble was never just **3**. It was **3 sixteens**. Cut out the sixteen cards on the lower half of the **card sheet**, take a set of singles, and come back.

In a hex pair, the left digit counts *cards* and the right digit counts *dots*.

34 — a number you have already met

$3 \text{ cards} = 48$
 $4 \text{ singles} = 4$
 $48 + 4 = 52$

Same number, third time. On page 2 it was 0011 0100, $32 + 16 + 4 = 52$. In hex, 34. On the desk, three cards and four singles. **Nothing changed but how you are looking at it.**

Your turn. Build each one, then fill the box in. **The 16 column is your cards; the 1 column is your singles** — same box as the nibble mats, place values on top, values underneath.

	16	1
Hex	4	B
Value		

Total = _____

	16	1
Hex	7	C
Value		

Total = _____

	16	1
Hex	D	E
Value		

Total = _____

You met DE on page 2. If the desk and the byte strip disagree, finding out which is wrong is the useful part.

Count out sixteen singles.

Now put them beside **one card**. They are worth exactly the same. **Sixteen singles are one card** — so swap them, and the right-hand digit drops to nought while the left-hand one goes up by one.

That swap is the only thing place value has ever meant.

Then stop using them.

You can build *every* byte there is — FF is 15 cards and 15 singles. But laying DE out takes a minute, and working it out takes five seconds: $13 \times 16 = 208$, plus **14**, is **222**.

The cards are how you learn to trust the arithmetic. Then you leave them behind.

Now put all sixteen cards on the desk at once. That is $16 \times 16 = 256$ dots — but a byte's biggest number is **255**, because FF is **15 cards and 15 singles** = 255. **The sixteenth card is one too many.** A byte counts **0 to 255** — 256 values, and you have just laid every one of them out. **Next: a colour code is three byte strips side by side.**